4.7 Types of methods

The different kinds of methods are:

`Constructor` — instantiates the class.

`Accessor` — accesses the member variables in the class.

`Mutator` — (also called Manipulator)—to change or write to the member variables of the class.

`Utility` — private methods that do work for the other methods of the class.

4.8 Our new class

Now we have a new class `Time1`. Class `Time1` keeps time variables. The class also validates any time we wish to create. It prevents us from creating an impossible time. As of yet, there is no actual object. Right now, it is only a recipe for an object of our type. In other words, we haven't instantiated it yet. We cannot instantiate it in this class. We need to create another class to do actually make an example of this class. We need to create a driver program `TimeTest.java`. The only purpose of this new class is to instantiate and test our new class `Time1`.

```
import javax.swing.JOptionPane;
public class TimeTest
{
public static void main( String args[] )
{
Time1 t = new Time1(); // calls Time1 constructor
}
}
```

Now we are using the class `Time1` that we just created. This line creates a reference "t" to an object of type Time1. Then, the "new Time1() calls the Constructor method to instantiate our new Time1 object "t".

```
import javax.swing.JOptionPane;
public class TimeTest
{
public static void main( String args[] )
{
Time1 t = new Time1(); // calls Time1 constructor
String output;
output = "The initial universal time is: " +
t.toUniversalString() + "\nThe initial time is: " +
```